

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15E

SPECTROSCOPY & NMR (CHROMATOGRAPHY, MASS SPEC, ^{13}C NMR, ^1H NMR & SPLITTING)

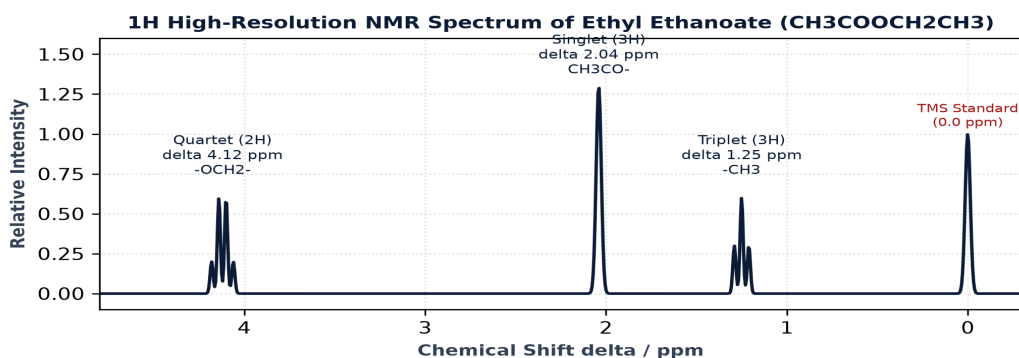
CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E: Spectroscopy & NMR (Chromatography, Mass Spec, ^{13}C NMR, ^1H NMR & Splitting)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	228 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. ¹H NMR Spectrum Analysis of Ethyl Ethanoate****[5 Marks]****WCH14/01/Jan23/Q23**

Ethyl ethanoate, CH₃COOCH₂CH₃, is analysed using high-resolution ¹H NMR spectroscopy.

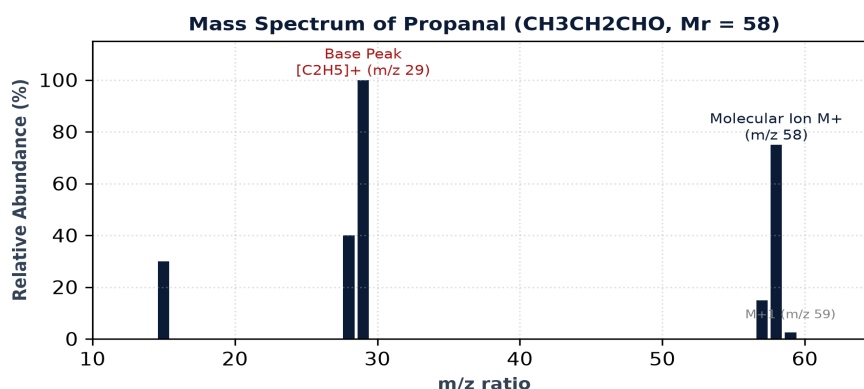


(a) Predict the number of peaks (environments) in the ¹H NMR spectrum of ethyl ethanoate. [1]

(b) Describe the splitting pattern (n+1 rule) and relative area for each peak. [4]

Q2. 2. Mass Spectrometry Fragmentation & D₂O Shake**[5 Marks]****WCH14/01/Oct22/Q24**

Compound X (C₃H₆O₂) has an IR absorption at 1715 cm⁻¹ and a broad absorption at 2500–3300 cm⁻¹.



(a) Identify the functional group present in X from the IR data. [1]

(b) State the effect of adding D₂O (D₂O shake) on the ¹H NMR spectrum of X. [2]

(c) Identify the fragment ion responsible for the peak at $m/z = 45$ in the mass spectrum of X. [2]

Q3. 3. Principles of Thin-Layer Chromatography (TLC)

[4 Marks]

WCH14/01/Jan22/Q15

A mixture of amino acids is separated using thin-layer chromatography (TLC) with a silica gel stationary phase and an organic solvent mobile phase.

(a) Define the term R_f value. [1]

(b) Explain why different amino acids have different R_f values on silica gel. [2]

(c) State how colorless amino acid spots are visualized on the TLC plate. [1]

Q4. 4. High-Performance Liquid Chromatography (HPLC)

[4 Marks]

WCH14/01/June21/Q16

HPLC is used to analyze pesticide residues in river water.

(a) Define retention time (t_R) in HPLC. [1]

(b) State two factors that affect the retention time of a compound in an HPLC column. [2]

(c) Explain how the concentration of a pesticide is determined from an HPLC chromatogram. [1]

Q5. 5. Gas Chromatography (GC) and Carrier Gases

[4 Marks]
WCH14/01/Jan21/Q17

Gas chromatography is used to separate volatile organic compounds.

(a) State the role of the mobile phase (carrier gas) in GC and name a suitable gas. [2]

(b) Explain why GC alone cannot definitively identify an unknown compound. [1]

(c) State how GC is combined with mass spectrometry (GC-MS) to overcome this limitation. [1]

Q6. 6. Mass Spectrometry: M^+ and $M+1$ Isotope Peaks

[4 Marks]
WCH14/01/Oct20/Q16

The mass spectrum of an unknown hydrocarbon shows a molecular ion peak M^+ at $m/z = 72$ and an $M+1$ peak at $m/z = 73$.

(a) The relative heights of the M^+ and $M+1$ peaks are 100 : 5.5. Calculate the number of carbon atoms in the molecule. [2]

(b) Deduce the molecular formula of the hydrocarbon. [1]

(c) State the natural isotope responsible for the M+1 peak. [1]

Q7. 7. Chlorine Isotope Ratio in Mass Spectrometry (M+ and M+2 Peaks)

[4 Marks]

WCH14/01/Jan20/Q17

The mass spectrum of a chloroalkane displays two molecular ion peaks at $m/z = 78$ and $m/z = 80$.

(a) State the natural abundance ratio of ^{35}Cl to ^{37}Cl isotopes. [1]

(b) Predict the relative peak height ratio of M+ ($m/z = 78$) to M+2 ($m/z = 80$). [1]

(c) Deduce the molecular formula of the chloroalkane. [2]

Q8. 8. Bromine Isotope Ratio in Mass Spectrometry

[4 Marks]

WCH14/01/June19/Q18

A bromoalkane sample exhibits two molecular ion peaks of equal intensity at $m/z = 122$ and $m/z = 124$.

(a) State the two naturally occurring isotopes of bromine and their relative abundance. [2]

(b) Explain why the peak heights at $m/z = 122$ and 124 are 1 : 1. [1]

(c) Deduce the structure of the bromoalkane. [1]

Q9. 9. Infrared Spectroscopy: Distinguishing Functional Groups**[4 Marks]**
WCH14/01/Jan19/Q15

Infrared spectra of three unknown liquids A, B, and C show distinct absorption bands:

Compound A: Broad absorption at 3350 cm^{-1} , no peak at 1700 cm^{-1} .

Compound B: Strong sharp absorption at 1715 cm^{-1} , no absorption above 3100 cm^{-1} .

Compound C: Very broad absorption at $2500\text{--}3300\text{ cm}^{-1}$ and strong peak at 1710 cm^{-1} .

(a) Identify the functional group present in Compound A. [1]

(b) Identify the functional group present in Compound B. [1]

(c) Identify the functional group present in Compound C. [1]

(d) State what bond causes the broad absorption in Compound A. [1]

Q10. 10. ^{13}C NMR Peak Counting & Molecular Symmetry**[4 Marks]**
WCH14/01/June18/Q15

Consider the four structural isomers of $\text{C}_4\text{H}_9\text{Cl}$:

1-chlorobutane, 2-chlorobutane, 1-chloro-2-methylpropane, 2-chloro-2-methylpropane.

(a) State the number of peaks expected in the ^{13}C NMR spectrum of 1-chlorobutane. [1]

(b) State the number of peaks expected in the ^{13}C NMR spectrum of 2-chloro-2-methylpropane. [1]

(c) Explain the difference in peak count between these two isomers in terms of symmetry. [2]

Q11. 11. Tetramethylsilane (TMS) Standard in NMR**[3 Marks]****WCH14/01/Jan18/Q14**

Tetramethylsilane, $\text{Si}(\text{CH}_3)_4$ (TMS), is added as an internal standard in ^1H and ^{13}C NMR spectroscopy.

- (a) Give two reasons why TMS is ideal as an NMR reference standard. [2]

- (b) State the chemical shift value assigned to TMS. [1]

Q12. 12. Solvents in ^1H NMR Spectroscopy and Deuterium Exchange**[4 Marks]****WCH14/01/Oct17/Q15**

Deuterated chloroform (CDCl_3) is commonly used as a solvent in ^1H NMR.

- (a) Explain why CDCl_3 is used rather than CHCl_3 as a solvent. [1]

- (b) State the effect of adding a drop of D_2O to a sample containing an alcohol before recording its ^1H NMR spectrum. [2]

- (c) Write the equation for the deuterium exchange reaction. [1]

Q13. 13. ^1H NMR Integration Traces**[3 Marks]****WCH14/01/June17/Q14**

The ^1H NMR spectrum of an unknown ester $\text{C}_3\text{H}_6\text{O}_2$ shows two signals with relative integration heights of 3 : 3.

- (a) State what information the integration trace provides in ^1H NMR. [1]

- (b) Deduce the structure of the ester $\text{C}_3\text{H}_6\text{O}_2$. [2]

Q14. 14. n+1 Rule for ^1H NMR Peak Splitting**[4 Marks]**
WCH14/01/Jan17/Q13

Apply the n+1 splitting rule to predict the ^1H NMR spectrum of 1,2-dichloroethane ($\text{ClCH}_2\text{CH}_2\text{Cl}$) versus 1,1-dichloroethane (CH_3CHCl_2).

(a) Predict the number of peaks and splitting pattern for 1,2-dichloroethane. [2]

(b) Predict the number of peaks and splitting patterns for 1,1-dichloroethane. [2]

Q15. 15. ^{13}C NMR Spectrum of Propanone vs Propanal**[3 Marks]**
WCH14/01/Oct16/Q14

Distinguish between propanone (CH_3COCH_3) and propanal ($\text{CH}_3\text{CH}_2\text{CHO}$) using ^{13}C NMR.

(a) State the number of peaks in the ^{13}C NMR spectrum of propanone. [1]

(b) State the number of peaks in the ^{13}C NMR spectrum of propanal. [1]

(c) Explain how the chemical shift of the carbonyl carbon distinguishes an aldehyde from a ketone. [1]

Q16. 16. Mass Spec Fragmentation of Propanal**[4 Marks]**
WCH14/01/June16/Q15

The mass spectrum of propanal ($\text{CH}_3\text{CH}_2\text{CHO}$, $M_r = 58$) exhibits prominent fragment ion peaks at $m/z = 57$, 29, and 15.

(a) Write the chemical formula and structure for the fragment ion at $m/z = 57$. [1]

(b) Write the chemical formula and structure for the fragment ion at $m/z = 29$. [2]

(c) Write the equation for the loss of a methyl radical from the molecular ion. [1]

Q17. 17. IR Absorption Bands of Alkenes vs Alkynes

[3 Marks]

WCH14/01/Jan16/Q16

Infrared spectra can identify unsaturated carbon-carbon bonds.

(a) State the wavenumber range for C=C double bond absorption. [1]

(b) State the wavenumber range for CC triple bond absorption. [1]

(c) Explain why symmetrical alkenes such as trans-hex-3-ene show no C=C absorption band in their IR spectrum. [1]

Q18. 18. Combined Spectroscopy: Formula C₂H₄O

[4 Marks]

WCH14/01/Oct15/Q15

An organic compound Z has empirical formula C₂H₄O and relative molecular mass 44.

(a) Deduce the molecular formula of Z. [1]

(b) Its IR spectrum shows a strong peak at 1720 cm⁻¹ and no peak above 3200 cm⁻¹. Its ¹H NMR spectrum shows a doublet (3H) and a quartet (1H). Deduce the structural formula of Z. [3]

[4 Marks]

Q19. 19. Spin-Spin Coupling Constants (J Values)

WCH14/01/June15/Q17

High-resolution ^1H NMR spectra display sub-peak splitting separated by coupling constants (J values).

(a) Explain what causes spin-spin coupling in ^1H NMR. [2]

(b) State why coupling constants between mutually split protons are equal ($J_{\text{AB}} = J_{\text{BA}}$). [1]

(c) Predict the appearance of a peak coupled to two non-equivalent neighboring protons with different J values. [1]

[3 Marks]

Q20. 20. Column Chromatography vs HPLC

WCH14/01/Jan15/Q14

Compare traditional gravity column chromatography with HPLC.

(a) State two advantages of HPLC over gravity column chromatography. [2]

(b) State why smaller particle sizes are used in HPLC column packing. [1]

[4 Marks]

Q21. 21. Mass Spec Fragmentation of Esters

WCH14/01/Oct14/Q16

The mass spectrum of methyl ethanoate ($\text{CH}_3\text{COOCH}_3$, $M_r = 74$) displays fragment peaks at $m/z = 59, 43, 31$, and 15 .

(a) Identify the ion responsible for the base peak at $m/z = 43$. [1]

(b) Identify the ion responsible for the peak at $m/z = 59$. [1]

(c) Identify the ion responsible for the peak at $m/z = 31$. [1]

(d) Define the term base peak in mass spectrometry. [1]

Q22. 22. ^1H NMR Spectrum of Propan-1-ol vs Propan-2-ol

[4 Marks]

WCH14/01/June14/Q18

Compare the ^1H NMR spectra of propan-1-ol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$) and propan-2-ol ($\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$).

(a) State the number of ^1H NMR peaks for propan-1-ol. [1]

(b) State the number of ^1H NMR peaks for propan-2-ol. [1]

(c) Predict the splitting pattern of the $-\text{CH}-$ proton peak in propan-2-ol (ignoring $-\text{OH}$ coupling). [2]

Q23. 23. Chromatography Retention Index & Kovats Index

[3 Marks]

WCH14/01/Jan14/Q15

Gas chromatography uses retention times and Kovats retention indices for compound identification.

(a) Explain why retention times vary if oven temperature fluctuates. [1]

(b) State how reference alkane standards are used to standardize retention indices. [2]

Q24. 24. IR Spectrum of Carboxylic Acid Anhydrides

[3 Marks]
WCH14/01/Oct13/Q16

Ethanoic anhydride displays two distinct C=O absorption peaks in its IR spectrum at 1760 cm⁻¹ and 1820 cm⁻¹.

(a) Explain why acid anhydrides exhibit a double C=O absorption peak. [2]

(b) Contrast this with the single C=O absorption peak of esters. [1]

Q25. 25. Mass Spec High-Resolution Mass Measurement

[4 Marks]
WCH14/01/June13/Q16

High-resolution mass spectrometry determines molecular masses to 4 decimal places.

Exact atomic masses: ¹H = 1.0078, ¹²C = 12.0000, ¹⁴N = 14.0031, ¹⁶O = 15.9949.

(a) Calculate the exact molecular mass of propanone (C₃H₆O). [1]

(b) Calculate the exact molecular mass of butane (C₄H₁₀). [1]

(c) Explain how high-resolution mass spectrometry distinguishes between propanone and butane, which both have nominal Mr = 58. [2]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. Multi-Spectral Structure Elucidation: Compound A (C₄H₈O₂)****[6 Marks]**
WCH14/01/Jan23/Q26

Compound A (C₄H₈O₂) is analysed by multiple spectroscopic techniques:

- Elemental Analysis: C = 54.55%, H = 9.09%, O = 36.36%.
- Mass Spectrum: M⁺ peak at m/z = 88, base peak at m/z = 43.
- IR Spectrum: Strong peak at 1740 cm⁻¹, no peak at 2500–3300 cm⁻¹ or 3200–3600 cm⁻¹.
- ¹H NMR: Singlet at 2.05 ppm (3H), Triplet at 1.25 ppm (3H), Quartet at 4.12 ppm (2H).

(a) Confirm the empirical formula of Compound A. [1]

(b) Deduce the functional group present from the IR spectrum. [1]

(c) Interpret each ¹H NMR peak to identify the proton environments and neighbors. [3]

(d) Draw the structural formula of Compound A and name it. [1]

Q27. 27. Multi-Spectral Structure Elucidation: Compound B (C₅H₁₀O)**[6 Marks]**
WCH14/01/Oct22/Q27

Compound B (C₅H₁₀O) gives a positive 2,4-DNPH test but negative Tollens' test.

- ¹³C NMR: 3 peaks at δ = 8.1, 35.2, 211.5 ppm.
- ¹H NMR: Triplet at 1.05 ppm (6H) and Quartet at 2.45 ppm (4H).

(a) Deduce the functional group of Compound B from chemical tests. [1]

(b) Explain why the ¹³C NMR spectrum shows only 3 peaks despite having 5 carbon atoms. [2]

(c) Interpret the ^1H NMR integration and splitting patterns. [2]

(d) Give the structural formula and systematic name of Compound B. [1]

Q28. 28. Multi-Spectral Structure Elucidation: Compound C ($\text{C}_4\text{H}_8\text{O}$)

[6 Marks]
WCH14/01/Jan22/Q25

Compound C ($\text{C}_4\text{H}_8\text{O}$) decolourises bromine water.

- IR Spectrum: Sharp absorption at 3620 cm^{-1} , strong peak at 1650 cm^{-1} .
- ^1H NMR: Doublet at 1.7 ppm (3H), Multiplet at 5.7 ppm (2H), Singlet at 3.2 ppm (1H), Doublet at 4.1 ppm (2H).

(a) Identify the two functional groups present in Compound C from chemical tests and IR. [2]

(b) State the effect of D_2O shake on the ^1H NMR spectrum. [1]

(c) Deduce the complete structure of Compound C. [3]

Q29. 29. ^{13}C NMR & ^1H NMR Structural Assignment of Isomeric Esters

[5 Marks]
WCH14/01/June21/Q26

Three structural isomers with formula $\text{C}_4\text{H}_8\text{O}_2$ are esters: Methyl propanoate, Ethyl ethanoate, Propyl methanoate.

(a) Predict the number of peaks in the ^{13}C NMR spectrum of each isomer. [2]

(b) Explain how the ^1H NMR spectrum of propyl methanoate ($\text{HCOOCH}_2\text{CH}_2\text{CH}_3$) distinguishes it from methyl propanoate ($\text{CH}_3\text{CH}_2\text{COOCH}_3$). [3]

Q30. 30. High-Resolution ^1H NMR Analysis of Ethanethiol

[5 Marks]

WCH14/01/Jan21/Q25

Ethanethiol ($\text{CH}_3\text{CH}_2\text{SH}$) is added to natural gas as an odorant.

(a) Predict the ^1H NMR spectrum of ethanethiol under high resolution assuming $-\text{SH}$ proton coupling occurs. [3]

(b) Describe how D_2O shake modifies the spectrum. [2]

Q31. 31. Mass Spectrometry Fragmentation Mechanisms of Carbonyls

[5 Marks]

WCH14/01/Oct20/Q25

Pentan-2-one ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$) undergoes alpha-cleavage and McLafferty rearrangement in mass spectrometry.

(a) Write equations for the two possible alpha-cleavage reactions of the molecular ion $[\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3]^+$. [3]

(b) State the m/z values of the resulting acylium fragment ions. [2]

Q32. 32. Chromatographic Separation & GC-MS Analysis of Essential Oils

[5 Marks]

WCH14/01/Jan20/Q26

Lavender oil contains linalool and linalyl acetate.

(a) Describe how GC-MS separates and quantifies linalool and linalyl acetate in lavender oil. [3]

(b) Explain how electronic mass spectral databases identify unknown peaks in the chromatogram. [2]

Q33. 33. ^{13}C NMR Chemical Shift Calculations and DEPT NMR

[5 Marks]

WCH14/01/June19/Q25

DEPT ^{13}C NMR spectroscopy distinguishes CH_3 , CH_2 , CH , and quaternary carbon atoms.

(a) Explain the appearance of CH_3 , CH_2 , and CH carbon signals in DEPT-135 NMR spectra. [3]

(b) Predict which carbon peak of 2-methylpropan-2-ol ($(\text{CH}_3)_3\text{COH}$) is ABSENT in DEPT-135. [2]

Q34. 34. Quantitative ^1H NMR Integration for Mixture Analysis**[6 Marks]**
WCH14/01/Jan19/Q26

A mixture contains ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) and ethyl ethanoate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$). The ^1H NMR spectrum shows:

- Peak 1: Singlet at 2.0 ppm ($\text{CH}_3\text{CO}-$ of ester) integration area = 12.0.
- Peak 2: Triplet at 1.2 ppm (CH_3 of ethanol) integration area = 8.0.

(a) Determine the relative moles of ethyl ethanoate in the sample from Peak 1. [2]

(b) Determine the relative moles of ethanol in the sample from Peak 2. [2]

(c) Calculate the mole percentage of ethanol in the mixture. [2]

Q35. 35. 2D NMR Spectroscopy Principles (COSY)**[5 Marks]**
WCH14/01/June18/Q26

2D ^1H - ^1H COSY (Correlation Spectroscopy) displays spin-spin coupling networks.

(a) Explain what off-diagonal cross-peaks represent in a 2D COSY spectrum. [2]

(b) Describe how COSY resolves overlapping multiplets in complex organic molecules. [3]

Q36. 36. Multi-Spectral Elucidation: Aromatic Derivative
C₈H₈O₂**[6 Marks]**
WCH14/01/Jan18/Q26

Compound D (C₈H₈O₂) contains a benzene ring.

- IR Spectrum: Sharp C=O at 1720 cm⁻¹, strong C-O at 1250 cm⁻¹.
- ¹H NMR: Singlet at 3.9 ppm (3H), Multiplet at 7.4–8.0 ppm (5H).

(a) Deduce the monosubstituted aromatic ring from ¹H NMR data. [1]

(b) Identify the group responsible for the 3.9 ppm singlet. [2]

(c) Deduce the structural formula and systematic name of Compound D. [3]

Q37. 37. ¹³C NMR Chemical Shift Trends in Substituted Benzenes**[5 Marks]**
WCH14/01/Oct17/Q26

Compare the ¹³C NMR spectra of 1,4-dimethylbenzene and 1,2-dimethylbenzene.

(a) Predict the number of ¹³C NMR peaks for 1,4-dimethylbenzene. [2]

(b) Predict the number of ¹³C NMR peaks for 1,2-dimethylbenzene. [2]

(c) Explain the difference based on symmetry elements. [1]

Q38. 38. Chiral Stationary Phases in Enantiomeric HPLC

[5 Marks]

WCH14/01/June17/Q26

Separation of racemic drug enantiomers requires Chiral Liquid Chromatography (CMP).

(a) Explain why conventional non-chiral HPLC stationary phases cannot separate enantiomers. [2]

(b) Describe how a chiral stationary phase (CSP) achieves separation of (R)- and (S)-enantiomers. [3]

Q39. 39. Mass Spec Fragmentation of Alcohols & M-18 Peak

[5 Marks]

WCH14/01/Jan17/Q25

Primary alcohols frequently show an $[M - 18]^+$ peak in their mass spectra.

(a) State the small molecule lost to produce the $[M - 18]^+$ peak. [1]

(b) Write the equation for the dehydration fragmentation of $[\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}]^+$. [2]

(c) Identify the ion structure formed at $m/z = 42$ for propan-1-ol ($M_r = 60$). [2]

Q40. 40. Infrared O-H Hydrogen Bonding Shift**[5 Marks]****WCH14/01/Oct16/Q25**

The IR spectrum of ethanol in gas phase shows a sharp O-H band at 3650 cm^{-1} , whereas liquid ethanol shows a broad band at 3350 cm^{-1} .

- (a) Explain why intermolecular hydrogen bonding broadens and shifts the O-H absorption to lower wavenumber in the liquid phase. [3]

- (b) Predict the effect of diluting liquid ethanol in non-polar CCl_4 solvent on the IR spectrum. [2]

Q41. 41. ^1H NMR Virtual Coupling and Complex Multiplets**[5 Marks]****WCH14/01/June16/Q26**

In 1-chloropropane ($\text{CH}_3\text{-CH}_2\text{-CH}_2\text{Cl}$), the central $\text{-CH}_2\text{-}$ protons appear as a sextet.

- (a) Apply the $n+1$ rule to explain why the central $\text{-CH}_2\text{-}$ protons form a sextet. [2]

- (b) State the assumption made about the coupling constants $J(\text{CH}_3\text{-CH}_2)$ and $J(\text{CH}_2\text{-CH}_2\text{Cl})$. [2]

- (c) Predict what happens if $J(\text{CH}_3\text{-CH}_2)$ is significantly different from $J(\text{CH}_2\text{-CH}_2\text{Cl})$. [1]

Q42. 42. Mass Spec High-Mass Resolution Isotope Patterns**[5 Marks]****WCH14/01/Jan16/Q25**

A compound containing two chlorine atoms displays three molecular ion peaks M^+ , $M+2$, and $M+4$.

(a) Calculate the theoretical relative peak height ratio for $M^+ : M+2 : M+4$ for a dichlorinated compound. [3]

(b) Deduce the peak height ratio for a dibrominated compound containing two bromine atoms. [2]

**Q43. 43. Multi-Spectral Structure Elucidation: Nitrogen
Compound C_3H_7NO**

[6 Marks]

WCH14/01/Oct15/Q26

Compound E (C_3H_7NO) is soluble in water.

- IR Spectrum: Strong $C=O$ at 1665 cm^{-1} , doublet peak at $3300\text{--}3400\text{ cm}^{-1}$.
- 1H NMR: Triplet at 1.1 ppm (3H), Quartet at 2.2 ppm (2H), Broad singlet at 6.5 ppm (2H).

(a) Identify the functional group from IR 1665 cm^{-1} and $3300\text{--}3400\text{ cm}^{-1}$ doublet. [2]

(b) Deduce the structure of Compound E from 1H NMR data. [3]

(c) Give the systematic IUPAC name for Compound E. [1]

**Q44. 44. HPLC Mass Spectrometry (LC-MS) Electrospray
Ionisation**

[5 Marks]

WCH14/01/June15/Q26

LC-MS uses soft electrospray ionisation (ESI) to analyze polar biomolecules.

(a) Explain why hard electron ionisation (EI) is unsuitable for thermally labile proteins. [2]

(b) Describe how ESI forms $[M + H]^+$ protonated pseudo-molecular ions. [2]

(c) State the m/z value observed for a peptide of mass 1200 Da forming a doubly charged $[M + 2H]^{2+}$ ion. [1]

Q45. 45. ^{13}C NMR DEPT & 2D HMQC Heteronuclear Correlation

[5 Marks]

WCH14/01/Jan15/Q26

HMQC (Heteronuclear Multiple Quantum Coherence) correlates ^1H chemical shifts with ^{13}C chemical shifts.

(a) State what a cross-peak in an HMQC spectrum indicates. [2]

(b) Explain how HMQC distinguishes between isomeric primary and secondary alcohols. [3]

Q46. 46. Supercritical Fluid Chromatography (SFC)

[5 Marks]

WCH14/01/Oct14/Q26

Supercritical fluid chromatography uses supercritical carbon dioxide (scCO_2) as mobile phase.

(a) Define a supercritical fluid. [1]

(b) State two operational advantages of SFC over HPLC. [2]

(c) Explain why scCO_2 is considered a green chromatography solvent. [2]

Q47. 47. ^1H NMR Anisotropic Shielding in Benzene & Carbonyls

[5 Marks]

WCH14/01/June14/Q27

Aromatic protons absorb at unusually low field (δ 6.5–8.5 ppm) due to magnetic anisotropy.

(a) Explain the origin of the ring current in benzene when placed in an external magnetic field B_0 . [2]

(b) Describe how the induced ring current magnetic field affects the effective magnetic field experienced by aromatic protons. [3]

Q48. 48. Mass Spec High-Resolution Fragmentation of Isomeric Ketones

[5 Marks]

WCH14/01/Jan14/Q26

Pentan-2-one and pentan-3-one both have molecular formula $\text{C}_5\text{H}_{10}\text{O}$ ($M_r = 86$).

(a) Predict the base peak m/z value for pentan-2-one resulting from α -cleavage. [2]

(b) Predict the base peak m/z value for pentan-3-one resulting from α -cleavage. [2]

(c) State how mass spectrometry unambiguously differentiates these two structural isomers. [1]

Q49. 49. Multi-Spectral Structural Elucidation: Complex Ester
C₆H₁₂O₂

[6 Marks]

WCH14/01/Oct13/Q27

Compound F (C₆H₁₂O₂) is hydrolysed by NaOH(aq) to give compound G and sodium ethanoate.

- ¹H NMR of F: Singlet at 2.0 ppm (3H), Doublet at 0.9 ppm (6H), Multiplet at 1.9 ppm (1H), Doublet at 3.8 ppm (2H).

(a) Identify the ethanoate part of the ester from ¹H NMR. [1]

(b) Deduce the structure of alcohol G formed on hydrolysis. [3]

(c) Draw the structural formula of Compound F and give its IUPAC name. [2]

Q50. 50. Master Structural Elucidation Challenge: Unknown
C₆H₁₀O₄

[6 Marks]

WCH14/01/June13/Q28

Compound X (C₆H₁₀O₄) is a symmetrical dicarboxylic acid ester used in polymer synthesis.

- IR Spectrum: C=O at 1735 cm⁻¹, C-O at 1180 cm⁻¹, no O-H peak.

- ¹³C NMR: 3 peaks at δ = 14.1, 60.5, 160.2 ppm.

- ¹H NMR: Triplet at 1.3 ppm (6H), Quartet at 4.3 ppm (4H).

(a) Deduce the number of ester groups present in Compound X. [1]

(b) Interpret the ^{13}C NMR and ^1H NMR signals to identify all carbon and proton environments. [3]

(c) Draw the complete displayed structure of Compound X and give its systematic name. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15E

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. ¹ H NMR Spectrum Analysis of Ethyl Ethanoate 1. (a) 3 peaks (3 proton environments) (1). 1. (b) CH ₃ CO- protons: singlet (0 adjacent H, n=0 → 1 peak), integration = 3H (1). -OCH ₂ - protons: quartet (3 adjacent H on CH ₃ , n=3 → 4 peaks), integration = 2H (1). -CH ₃ protons of ethyl: triplet (2 adjacent H on CH ₂ , n=2 → 3 peaks), integration = 3H (1).	[5]
Q2	Q2: 2. Mass Spectrometry Fragmentation & D ₂ O Shake 2. (a) Carboxylic acid (-COOH) group (C=O at 1715, O-H at 2500–3300) (1). 2. (b) The acidic -OH proton exchanges with deuterium (-OH + D ₂ O → -OD + HOD); the -OH peak at delta ~11 ppm disappears (2). 2. (c) [COOH] ⁺ ion (m/z = 12 + 32 + 1 = 45) (2).	[5]
Q3	Q3: 3. Principles of Thin-Layer Chromatography (TLC) 3. (a) R _f = (distance moved by spot) / (distance moved by solvent front) (1). 3. (b) Different amino acids have different relative polarities; more polar amino acids adsorb more strongly to the polar silica stationary phase and travel slower (1); less polar amino acids dissolve more readily in the mobile phase and travel faster (1). 3. (c) Spray plate with ninhydrin reagent and heat (forms purple spots) or view under UV light (1).	[4]
Q4	Q4: 4. High-Performance Liquid Chromatography (HPLC) 4. (a) Retention time is the time taken for a compound to travel from injection to detection at the column outlet (1). 4. (b) Polarity of the stationary/mobile phase (1); flow rate of mobile phase / column temperature (1). 4. (c) The area under the chromatographic peak is proportional to compound concentration (measured against a calibration curve) (1).	[4]
Q5	Q5: 5. Gas Chromatography (GC) and Carrier Gases 5. (a) Mobile phase carries the gaseous sample through the column (1); inert gas such as helium (He) or nitrogen (N ₂) (1). 5. (b) Different compounds can have identical retention times under specific conditions (1). 5. (c) Separated components exiting GC pass directly into a mass spectrometer to generate unique fragmentation mass spectra (GC-MS) (1).	[4]
Q6	Q6: 6. Mass Spectrometry: M ⁺ and M+1 Isotope Peaks 6. (a) Number of C atoms = (height M+1 / height M ⁺) × (100 / 1.1) (1) = (5.5 / 100) × (100 / 1.1) = 5 carbon atoms (1). 6. (b) C ₅ H ₁₂ (Pentane isomers, M _r = (5×12)+(12×1) = 72) (1). 6. (c) Carbon-13 isotope (¹³ C) (1).	[4]
Q7	Q7: 7. Chlorine Isotope Ratio in Mass Spectrometry (M ⁺ and M+2 Peaks) 7. (a) 3 : 1 ratio of ³⁵ Cl : ³⁷ Cl (1). 7. (b) 3 : 1 relative peak height ratio (1). 7. (c) C ₃ H ₇ Cl (Chloropropane, M _r with ³⁵ Cl = (3×12)+7+35 = 78) (2).	[4]
Q8	Q8: 8. Bromine Isotope Ratio in Mass Spectrometry 8. (a) ⁷⁹ Br and ⁸¹ Br (1); 50 : 50 (1 : 1) natural abundance ratio (1). 8. (b) Equal probability of a molecule containing ⁷⁹ Br (m/z 122) or ⁸¹ Br (m/z 124) (1). 8. (c) CH ₃ CH ₂ CH ₂ Br or CH ₃ CH(Br)CH ₃ (Bromopropane, M _r = 43 + 79 = 122) (1).	[4]
Q9	Q9: 9. Infrared Spectroscopy: Distinguishing Functional Groups 9. (a) Alcohol (-OH) group (1). 9. (b) Carbonyl group (Aldehyde or Ketone C=O) (1). 9. (c) Carboxylic acid (-COOH) group (1). 9. (d) O-H stretch involved in hydrogen bonding (1).	[4]

Q10	Q10: 10. ¹³ C NMR Peak Counting & Molecular Symmetry 10. (a) 4 peaks (4 non-equivalent carbon environments) (1). 10. (b) 2 peaks (2 non-equivalent carbon environments) (1). 10. (c) 2-chloro-2-methylpropane (CH ₃) ₃ CCl has high molecular symmetry where all 3 methyl carbons are chemically equivalent (1); 1-chlorobutane has no internal symmetry plane, making all 4 carbon atoms chemically distinct (1).	[4]
Q11	Q11: 11. Tetramethylsilane (TMS) Standard in NMR 11. (a) Gives a single sharp peak because all 12 H atoms / 4 C atoms are identical (1); inert, non-toxic, and volatile (bp 27°C) so easily removed (1). 11. (b) 0.0 ppm (1).	[3]
Q12	Q12: 12. Solvents in ¹ H NMR Spectroscopy and Deuterium Exchange 12. (a) Deuterium (² H) does not absorb radiofrequency radiation in the ¹ H NMR range, avoiding solvent peak interference (1). 12. (b) The -OH proton peak disappears from the spectrum (2). 12. (c) R-OH + D ₂ O → R-OD + HOD (1).	[4]
Q13	Q13: 13. ¹ H NMR Integration Traces 13. (a) Integration trace area is proportional to the relative number of hydrogen atoms in that environment (1). 13. (b) Methyl ethanoate, CH ₃ COOCH ₃ (1); contains two separate methyl groups each with 3 equivalent H atoms (1).	[3]
Q14	Q14: 14. n+1 Rule for ¹ H NMR Peak Splitting 14. (a) 1,2-dichloroethane: 1 peak (singlet) because all 4 protons are chemically equivalent due to symmetry (2). 14. (b) 1,1-dichloroethane: 2 peaks; CH ₃ protons appear as a doublet (1 adjacent H, n=1 → 2 sub-peaks) (1); -CH- proton appears as a quartet (3 adjacent H, n=3 → 4 sub-peaks) (1).	[4]
Q15	Q15: 15. ¹³ C NMR Spectrum of Propanone vs Propanal 15. (a) 2 peaks (symmetrical molecule) (1). 15. (b) 3 peaks (all 3 carbons non-equivalent) (1). 15. (c) Aldehyde C=O appears at delta 190-200 ppm; ketone C=O appears at delta 200-220 ppm (1).	[3]
Q16	Q16: 16. Mass Spec Fragmentation of Propanal 16. (a) [CH ₃ CH ₂ CO] ⁺ or [M - H] ⁺ (loss of aldehyde H) (1). 16. (b) [CHO] ⁺ or [C ₂ H ₅] ⁺ ion (2). 16. (c) [CH ₃ CH ₂ CHO] ⁺ → [CH ₂ CHO] ⁺ + .CH ₃ (1).	[4]
Q17	Q17: 17. IR Absorption Bands of Alkenes vs Alkynes 17. (a) 1620 - 1680 cm ⁻¹ (1). 17. (b) 2100 - 2260 cm ⁻¹ (1). 17. (c) IR absorption requires a change in dipole moment during bond vibration; symmetrical trans-hex-3-ene has zero net dipole change upon stretching (1).	[3]
Q18	Q18: 18. Combined Spectroscopy: Formula C ₂ H ₄ O 18. (a) C ₂ H ₄ O (Mr = (2×12)+(4×1)+16 = 44) (1). 18. (b) IR 1720 cm ⁻¹ confirms C=O carbonyl (1); ¹ H NMR doublet (3H) = CH ₃ group next to 1 H (1); quartet (1H) = -CHO aldehyde proton next to CH ₃ (1); Structure = CH ₃ CHO (ethanal) (1).	[4]
Q19	Q19: 19. Spin-Spin Coupling Constants (J Values) 19. (a) Interaction between the magnetic moments of adjacent non-equivalent protons transmitted through chemical bonds (2). 19. (b) Both proton environments interact with the same coupling energy across the intervening C-C bond (1). 19. (c) Doublet of doublets (4 sub-peaks of equal height) (1).	[4]
Q20	Q20: 20. Column Chromatography vs HPLC 20. (a) Higher resolution / faster separation speed (1); automated high-sensitivity detection and quantitative analysis (1). 20. (b) Smaller particle sizes increase surface area of stationary phase, improving separation efficiency (1).	[3]

Q21	Q21: 21. Mass Spec Fragmentation of Esters 21. (a) $[\text{CH}_3\text{CO}]^+$ (acetyl ion) (1). 21. (b) $[\text{COOCH}_3]^+$ or $[\text{M} - \text{CH}_3]^+$ (1). 21. (c) $[\text{OCH}_3]^+$ (methoxy ion) (1). 21. (d) The base peak is the most intense peak in the mass spectrum, assigned 100% relative abundance (1).	[4]
Q22	Q22: 22. ^1H NMR Spectrum of Propan-1-ol vs Propan-2-ol 22. (a) 4 peaks (4 proton environments) (1). 22. (b) 3 peaks (2 symmetrical methyl groups = 6H, 1 CH = 1H, 1 OH = 1H) (1). 22. (c) Septet (7 sub-peaks) (1); adjacent to 6 equivalent H atoms on the two CH_3 groups ($n=6 \rightarrow 6+1 = 7$) (1).	[4]
Q23	Q23: 23. Chromatography Retention Index & Kovats Index 23. (a) Higher temperature increases vapour pressure, decreasing retention time on the stationary phase (1). 23. (b) Sample retention time is compared against unbranched alkanes injected under identical conditions (1); Kovats index interpolates relative retention between alkane standards (1).	[3]
Q24	Q24: 24. IR Spectrum of Carboxylic Acid Anhydrides 24. (a) Vibrational coupling between the two carbonyl $\text{C}=\text{O}$ bonds produces asymmetric stretching (higher frequency $\sim 1820 \text{ cm}^{-1}$) and symmetric stretching (lower frequency $\sim 1760 \text{ cm}^{-1}$) (2). 24. (b) Esters contain only one $\text{C}=\text{O}$ group per functional unit, giving a single sharp absorption peak at $1735\text{--}1750 \text{ cm}^{-1}$ (1).	[3]
Q25	Q25: 25. Mass Spec High-Resolution Mass Measurement 25. (a) $\text{C}_3\text{H}_6\text{O}$: $3(12.0000) + 6(1.0078) + 15.9949 = 58.0417$ (1). 25. (b) C_4H_{10}: $4(12.0000) + 10(1.0078) = 58.0780$ (1). 25. (c) High-resolution MS measures M^+ to 4 decimal places; 58.0417 is uniquely assigned to $\text{C}_3\text{H}_6\text{O}$ while 58.0780 is assigned to C_4H_{10} (2).	[4]
Q26	Q26: 26. Multi-Spectral Structure Elucidation: Compound A ($\text{C}_4\text{H}_8\text{O}_2$) 26. (a) C: $54.55/12 = 4.546$; H: $9.09/1 = 9.09$; O: $36.36/16 = 2.272$. Ratio 2:4:1 $\rightarrow$ Empirical $\text{C}_2\text{H}_4\text{O}$ (1). 26. (b) $\text{C}=\text{O}$ absorption at 1740 cm^{-1} with no O-H stretch confirms an ester group (1). 26. (c) Singlet at 2.05 ppm (3H) = $\text{CH}_3\text{-CO-}$ group (1). Quartet at 4.12 ppm (2H) and triplet at 1.25 ppm (3H) = $\text{-O-CH}_2\text{-CH}_3$ ethyl group (2). 26. (d) $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (Ethyl ethanoate) (1).	[6]
Q27	Q27: 27. Multi-Spectral Structure Elucidation: Compound B ($\text{C}_5\text{H}_{10}\text{O}$) 27. (a) Positive 2,4-DNPH confirms carbonyl $\text{C}=\text{O}$; negative Tollens' test rules out aldehyde $\rightarrow$ Ketone (1). 27. (b) Symmetrical ketone structure creates identical carbon environments on both sides of $\text{C}=\text{O}$ (2). 27. (c) Triplet at 1.05 ppm (6H) = 2 equivalent CH_3 groups split by adjacent CH_2 (1); Quartet at 2.45 ppm (4H) = 2 equivalent $\text{-CH}_2\text{-}$ groups next to $\text{C}=\text{O}$ split by CH_3 (1). 27. (d) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ (Pentan-3-one) (1).	[6]
Q28	Q28: 28. Multi-Spectral Structure Elucidation: Compound C ($\text{C}_4\text{H}_8\text{O}$) 28. (a) Decolourises bromine water & IR 1650 cm^{-1} = Alkene $\text{C}=\text{C}$ (1); IR 3620 cm^{-1} = Alcohol -OH (1). 28. (b) Singlet at 3.2 ppm (1H) disappears (1). 28. (c) $\text{CH}_3\text{-CH=CH-CH}_2\text{OH}$ (But-2-en-1-ol) (3).	[6]
Q29	Q29: 29. ^{13}C NMR & ^1H NMR Structural Assignment of Isomeric Esters 29. (a) Methyl propanoate: 4 peaks (1); Ethyl ethanoate: 4 peaks; Propyl methanoate: 4 peaks (1). 29. (b) Propyl methanoate has a unique low-field singlet at $\delta \sim 8.0$ ppm for the formate proton (HCOO-) (1); Propyl methanoate has 4 proton environments giving a singlet (1H), triplet (2H), multiplet (2H), triplet (3H) (1); Methyl propanoate has 3 proton environments giving a singlet (3H), quartet (2H), triplet (3H) (1).	[5]

Q30	Q30: 30. High-Resolution ^1H NMR Analysis of Ethanethiol 30. (a) CH_3 protons: triplet (3H) at δ 1.3 ppm (1); $-\text{CH}_2-$ protons: doublet of quartets (2H) at δ 2.5 ppm split by CH_3 and SH (1); $-\text{SH}$ proton: triplet (1H) at δ 1.5 ppm split by CH_2 (1). 30. (b) The $-\text{SH}$ triplet proton peak disappears due to deuterium exchange ($-\text{SH} + \text{D}_2\text{O} \rightarrow -\text{SD} + \text{HOD}$) (1); the $-\text{CH}_2-$ peak simplifies from a complex multiplet to a clean quartet (1).	[5]
Q31	Q31: 31. Mass Spectrometry Fragmentation Mechanisms of Carbonyls 31. (a) Cleavage 1: $[\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3]^+ \rightarrow [\text{CH}_3\text{CO}]^+ (m/z\ 43) + \cdot\text{CH}_2\text{CH}_2\text{CH}_3$ (1.5). Cleavage 2: $[\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3]^+ \rightarrow [\text{COCH}_2\text{CH}_2\text{CH}_3]^+ (m/z\ 71) + \cdot\text{CH}_3$ (1.5). 31. (b) $m/z = 43$ ($[\text{CH}_3\text{CO}]^+$ base peak) (1); $m/z = 71$ ($[\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}]^+$) (1).	[5]
Q32	Q32: 32. Chromatographic Separation & GC-MS Analysis of Essential Oils 32. (a) Sample vaporised and carried by He gas through a capillary column coated with stationary phase (1); linalool and linalyl acetate elute at distinct retention times t_R (1); detector measures peak areas proportional to relative concentrations (1). 32. (b) Mass spectrum generated at peak apex is compared against a digital library of reference fragmentation patterns (1); match factor / similarity index confirms chemical identity (1).	[5]
Q33	Q33: 33. ^{13}C NMR Chemical Shift Calculations and DEPT NMR 33. (a) CH_3 and CH carbon signals point UP (positive phase) (1); CH_2 carbon signals point DOWN (negative phase) (1); Quaternary carbon signals (no H attached) are completely ABSENT (1). 33. (b) Central quaternary carbon (C-2 attached to $-\text{OH}$ and 3 CH_3 groups) has zero attached protons and will be ABSENT in DEPT-135 (2).	[5]
Q34	Q34: 34. Quantitative ^1H NMR Integration for Mixture Analysis 34. (a) Peak 1 singlet represents 3H of $\text{CH}_3\text{CO}-$ in ester. Relative moles ester = $12.0 / 3 = 4.0$ mol units (2). 34. (b) Peak 2 triplet represents 3H of CH_3 in ethanol. Relative moles ethanol = $8.0 / 3 = 2.67$ mol units (2). 34. (c) Total mole units = $4.0 + 2.67 = 6.67$; Mole % ethanol = $(2.67 / 6.67) \times 100 = 40.0\%$ (2).	[6]
Q35	Q35: 35. 2D NMR Spectroscopy Principles (COSY) 35. (a) Cross-peaks off the diagonal indicate scalar spin-spin coupling between two specific proton environments (2). 35. (b) Tracing horizontal and vertical lines from a cross-peak directly identifies which proton peaks are adjacent/coupled (2), bypassing 1D multiplet overlap (1).	[5]
Q36	Q36: 36. Multi-Spectral Elucidation: Aromatic Derivative $\text{C}_8\text{H}_8\text{O}_2$ 36. (a) 5H multiplet at 7.4–8.0 ppm confirms a monosubstituted benzene ring (C_6H_5-) (1). 36. (b) Singlet at 3.9 ppm (3H) attached to oxygen = $-\text{OCH}_3$ methoxy ester group (2). 36. (c) $\text{C}_6\text{H}_5\text{COOCH}_3$ (Methyl benzoate) (3).	[6]
Q37	Q37: 37. ^{13}C NMR Chemical Shift Trends in Substituted Benzenes 37. (a) 3 peaks (methyl C, ipso aromatic C, ortho aromatic C) (2). 37. (b) 4 peaks (methyl C, ipso C-1/2, C-3/6, C-4/5) (2). 37. (c) 1,4-isomer has two perpendicular planes of symmetry; 1,2-isomer has only one plane of symmetry (1).	[5]
Q38	Q38: 38. Chiral Stationary Phases in Enantiomeric HPLC 38. (a) Enantiomers have identical physical properties (boiling point, solubility, polarity) in an achiral environment (2). 38. (b) A chiral stationary phase contains single-enantiomer selectors (3D chiral cavities) (1); (R)- and (S)-enantiomers form diastereomeric transient complexes with different stability constants (1), resulting in different retention times t_R (1).	[5]
Q39	Q39: 39. Mass Spec Fragmentation of Alcohols & M-18 Peak 39. (a) Water molecule (H_2O, mass 18) (1). 39. (b) $[\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}]^+ \rightarrow [\text{CH}_3\text{CH}=\text{CH}_2]^+ (m/z\ 42) + \text{H}_2\text{O}$ (2). 39. (c) Propene radical cation $[\text{C}_3\text{H}_6]^+ (m/z = 42)$ (2).	[5]

Q40	Q40: 40. Infrared O–H Hydrogen Bonding Shift 40. (a) Hydrogen bonding weakens the O–H covalent bond by pulling electron density away (1), reducing the force constant of the O–H stretch (1); variable strength H-bonds across liquid molecules create a wide distribution of vibrational frequencies (broad peak) (1). 40. (b) Dilution separates ethanol molecules, breaking H-bonds; the broad 3350 cm ⁻¹ band diminishes while a sharp free O–H band at 3650 cm ⁻¹ appears (2).	[5]
Q41	Q41: 41. 1H NMR Virtual Coupling and Complex Multiplets 41. (a) Central –CH2– is adjacent to 3 protons on CH3 and 2 protons on CH2Cl (total n = 3+2 = 5) (1); n+1 = 5+1 = 6 sub-peaks (sextet) (1). 41. (b) Assumes both coupling constants are approximately equal (J1 ≈ J2) (2). 41. (c) Peak splits into a complex triplet of quartets (12 sub-peaks) (1).	[5]
Q42	Q42: 42. Mass Spec High-Mass Resolution Isotope Patterns 42. (a) ³⁵ Cl : ³⁷ Cl = 3 : 1. Ratio = (3+1) ² = 3 ² : 2(3x1) : 1 ² = 9 : 6 : 1 (3). 42. (b) ⁷⁹ Br : ⁸¹ Br = 1 : 1. Ratio = (1+1) ² = 1 ² : 2(1x1) : 1 ² = 1 : 2 : 1 (2).	[5]
Q43	Q43: 43. Multi-Spectral Structure Elucidation: Nitrogen Compound C ₃ H ₇ NO 43. (a) Primary amide group –CONH ₂ (C=O at 1665, N–H doublet at 3300–3400 cm ⁻¹) (2). 43. (b) Triplet 1.1 ppm (3H) + Quartet 2.2 ppm (2H) = ethyl group CH ₃ CH ₂ – (2); Broad singlet 6.5 ppm (2H) = –CONH ₂ protons (1). Structure = CH ₃ CH ₂ CONH ₂ (1). 43. (c) Propanamide (1).	[6]
Q44	Q44: 44. HPLC Mass Spectrometry (LC–MS) Electrospray Ionisation 44. (a) EI causes extensive fragmentation and thermal decomposition of fragile non-volatile molecules, destroying the molecular ion (2). 44. (b) High voltage applied to nebulized aerosol adds H ⁺ protons to basic sites, forming intact [M + H] ⁺ ions without fragmenting (2). 44. (c) m/z = (1200 + 2) / 2 = 601.0 (1).	[5]
Q45	Q45: 45. ¹³ C NMR DEPT & 2D HMQC Heteronuclear Correlation 45. (a) Shows direct 1-bond coupling between a specific proton and the carbon atom to which it is attached (2). 45. (b) Primary alcohol: –CH ₂ OH proton signal (~3.6 ppm) correlates with ¹³ C peak at ~60 ppm (1); Secondary alcohol: –CH(OH)– proton signal (~3.8 ppm) correlates with ¹³ C peak at ~68 ppm (1); confirms exact C–H pairing (1).	[5]
Q46	Q46: 46. Supercritical Fluid Chromatography (SFC) 46. (a) A substance above its critical temperature and pressure exhibiting gas-like diffusivity and liquid-like solvent power (1). 46. (b) Faster flow rates due to lower viscosity (1); rapid non-thermal sample recovery by depressurising CO ₂ gas (1). 46. (c) Non-toxic, non-flammable, naturally abundant, and leaves zero organic solvent residue upon evaporation (2).	[5]
Q47	Q47: 47. 1H NMR Anisotropic Shielding in Benzene & Carbonyls 47. (a) Delocalised pi-electrons circulate around the benzene ring, creating an induced ring current (2). 47. (b) The induced magnetic field opposes B ₀ inside the ring but reinforces B ₀ outside the ring where aromatic H atoms reside (2); protons experience a higher effective magnetic field and are deshielded, shifting downfield to 6.5–8.5 ppm (1).	[5]
Q48	Q48: 48. Mass Spec High-Resolution Fragmentation of Isomeric Ketones 48. (a) [CH ₃ CO] ⁺ ion at m/z = 43 (cleavage of C1–C2 bond) (2). 48. (b) [CH ₃ CH ₂ CO] ⁺ ion at m/z = 57 (cleavage of C2–C3 bond) (2). 48. (c) The different m/z values of their acylium fragment base peaks (43 vs 57) uniquely identify each isomer (1).	[5]
Q49	Q49: 49. Multi-Spectral Structural Elucidation: Complex Ester C ₆ H ₁₂ O ₂ 49. (a) Singlet at 2.0 ppm (3H) = CH ₃ CO– ethanoate group (1). 49. (b) Doublet 0.9 ppm (6H) + Multiplet 1.9 ppm (1H) + Doublet 3.8 ppm (2H) = 2-methylpropan-1-ol (CH ₃) ₂ CHCH ₂ OH (3). 49. (c) CH ₃ COOCH ₂ CH(CH ₃) ₂ (2-methylpropyl ethanoate / isobutyl ethanoate) (2).	[6]

Q50	Q50: 50. Master Structural Elucidation Challenge: Unknown $C_6H_{10}O_4$ 50. (a) Formula $C_6H_{10}O_4$ contains 4 oxygens and 2 degrees of unsaturation $\rightarrow$ Diester (2 ester groups) (1). 50. (b) ^{13}C NMR 3 peaks confirms high symmetry (1); 1H NMR triplet (6H) + quartet (4H) = 2 equivalent ethyl groups ($-OCH_2CH_3$) attached to oxygen (2). 50. (c) $CH_3CH_2OOC-COOCH_2CH_3$ (Diethyl ethanedioate / diethyl oxalate) (2).	[6]
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SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15E**FAQ #1: n+1 Splitting Rule Application****Category: 1H NMR Splitting****COMMON EXAMINER TRAP / MISTAKE:**

Counting protons on the same carbon atom when determining splitting.

MODEL ANSWER / EXAMINER EXPECTATION:

The n+1 rule states that a proton peak is split into n+1 sub-peaks, where n is the number of protons on ADJACENT carbon atoms (3 bonds away). Protons on the same carbon do not split each other.

FAQ #2: D2O Exchange Mechanism**Category: D2O Shake****COMMON EXAMINER TRAP / MISTAKE:**

Expecting C-H protons to exchange with D2O.

MODEL ANSWER / EXAMINER EXPECTATION:

D2O shake ONLY removes peaks for labile -OH and -NH protons because they exchange rapidly with deuterium (-OH + D2O → -OD + HOD). Protons attached directly to carbon (C-H) do NOT exchange.

FAQ #3: TMS Reference Peak**Category: NMR Standard****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting the chemical shift value of TMS.

MODEL ANSWER / EXAMINER EXPECTATION:

Tetramethylsilane Si(CH₃)₄ (TMS) is assigned a chemical shift of EXACTLY 0.0 ppm. Its 12 identical H atoms give a sharp reference peak at the far right of the spectrum.

FAQ #4: 13C NMR Integration Traces**Category: 13C NMR vs 1H NMR****COMMON EXAMINER TRAP / MISTAKE:**

Assuming peak heights in 13C NMR represent the number of carbons.

MODEL ANSWER / EXAMINER EXPECTATION:

13C NMR peak heights/areas are NOT strictly proportional to the number of carbon atoms (unlike 1H NMR integration). ONLY the number of peaks (carbon environments) and chemical shift values are evaluated.

FAQ #5: Mass Spec Molecular Ion Peak Identification**Category: Mass Spectrometry****COMMON EXAMINER TRAP / MISTAKE:**

Confusing the molecular ion M⁺ peak with fragment peaks.

MODEL ANSWER / EXAMINER EXPECTATION:

The molecular ion M⁺ peak is the peak with the HIGHEST m/z value (excluding small M+1 isotope peaks). Its m/z value gives the relative molecular mass (Mr) of the compound.

FAQ #6: Mass Spec Fragment Ion Positive Charge Requirement**Category: Mass Spec Formulas****COMMON EXAMINER TRAP / MISTAKE:**

Writing fragment species without a positive charge.

MODEL ANSWER / EXAMINER EXPECTATION:

Mass spectrometers ONLY detect POSITIVELY CHARGED ions. Fragment formulas MUST include a positive charge (e.g. $[\text{CH}_3\text{CO}]^+$, $[\text{C}_2\text{H}_5]^+$, $[\text{COOH}]^+$).

FAQ #7: Infrared O–H Absorption Differences**Category: IR Spectroscopy****COMMON EXAMINER TRAP / MISTAKE:**

Confusing alcohol O–H with carboxylic acid O–H absorption.

MODEL ANSWER / EXAMINER EXPECTATION:

Alcohol O–H absorbs as a smooth broad peak at $3200\text{--}3600\text{ cm}^{-1}$. Carboxylic acid O–H absorbs as a VERY BROAD, distorted peak at $2500\text{--}3300\text{ cm}^{-1}$ overlapping the C–H region.

FAQ #8: Deuterated Solvents in ^1H NMR**Category: NMR Solvents****COMMON EXAMINER TRAP / MISTAKE:**

Using H_2O or CHCl_3 as an NMR solvent.

MODEL ANSWER / EXAMINER EXPECTATION:

Solvents like H_2O or CHCl_3 contain ^1H protons that would produce huge peaks masking the sample. Deuterated solvents like CDCl_3 or D_2O are used because deuterium (^2H) does not absorb in the ^1H frequency range.

FAQ #9: Singlets for –OH and –NH Protons**Category: ^1H NMR Features****COMMON EXAMINER TRAP / MISTAKE:**

Expecting –OH protons to show splitting from adjacent C–H protons.

MODEL ANSWER / EXAMINER EXPECTATION:

Protons on –OH and –NH groups undergo rapid exchange with trace water/solvents, averaging out spin–spin coupling. Consequently, –OH and –NH protons almost always appear as SINGLETs.

FAQ #10: GC–MS Hyphenated Technique Function**Category: Chromatography–MS****COMMON EXAMINER TRAP / MISTAKE:**

Claiming GC identifies compounds without MS.

MODEL ANSWER / EXAMINER EXPECTATION:

Gas Chromatography (GC) SEPARATES the components of a complex mixture based on retention time. Mass Spectrometry (MS) IDENTIFIES each separated component by its mass spectrum fingerprint.